Layer 7: SAM Chemistry

Three-Component Probe-Thiol/C6FC/MCH Monolayer: Deposition, Stability, and Ca^{2+} Artefact

Eton iGEM 2026, Dry Lab | Layer 7 of 17-Layer Biosensor Pipeline | `sam_loading.py`, `sam_stability.py`, `calcium_artefact.py`

Layer 7 characterises the self-assembled monolayer (SAM) that immobilises the peptide probe on the gold electrode surface. Its primary output is the probe surface density Γ_p, which passes to Layer 8 (steric accessibility) and Layer 9 (surface kinetics). Secondary outputs are bounds on SAM degradation over the 60-minute assay window and a quantitative account of the Ca^{2+} artefact and its ratiometric cancellation.

Every parameter value must be confirmed from a paper physically present in `papers/layer_07/`; values without a paper source are explicitly flagged throughout.

I. Mixed SAM Architecture

Molecular components

The sensor electrode carries a three-component thiol monolayer. The probe-thiol consists of a peptide substrate bearing a methylene-blue (MB) redox label, tethered to gold via a C_6 alkyl spacer and a PEG_6 flexible linker (HS–$(CH_2)_6$–PEG_6–SGKIPRTLTA–MB). MMP-9 (or HNE on the secondary channel) cleaves the peptide, releasing the MB-bearing fragment into solution and producing a signal-off decrease in SWV peak current proportional to enzymatic activity (Layer 0, Section V).

A co-immobilised reference reporter, 6-ferrocenylhexanethiol (C6FC, Fc–$(CH_2)_6$–SH), provides an internal electrochemical standard. C6FC shares the same C_6 alkyl chain as the probe-thiol spacer, ensuring compatible packing geometry in the SAM. The ferrocene (Fc) headgroup gives a reversible one-electron redox signal. Adjémian et al. (2010) [12] report a formal potential of ∼150–155 mV vs. SCE (≈ +0.20 V vs. Ag/AgCl) for an alkyl-ferrocene label on an analogous Fc-peptide-PEG monolayer on gold (p. 10350 of [12]); this value is expected to be close to the C6FC formal potential but must be confirmed directly on the fabricated device. The Fc signal is well positive of the MB reduction peak; the two peaks are resolved in a single SWV scan without additional switching (Section I.3). The MB formal potential in the assay buffer is not confirmed from any paper in `papers/layer_07/` and must be determined by cyclic voltammetry on the fabricated electrode.

The diluent is 6-mercapto-1-hexanol (MCH, HS–$(CH_2)_6$–OH). MCH fills gold sites not occupied by the probe-thiol or C6FC, presenting a chemically inert surface to the wound fluid and preventing non-specific adsorption of wound proteins to the bare gold.

Close-packed monolayer density

The maximum surface density of a fully close-packed alkanethiol monolayer on gold, measured by reductive desorption coulometry, is [6]:

$$\Gamma_{\max} = 9.3 \times 10^{-10} \text{ mol cm}^{-2} = 930 \text{ pmol cm}^{-2} \tag{I.1}$$

Love et al. report an independent cross-check of approximately 750 pmol cm^{-2} for densely packed alkanethiol monolayers on Au(111) [1]. Equation (I.1) is an upper bound. The probe-thiol cannot approach $\Gamma_{\max}$ due to the bulky peptide headgroup; the target probe coverage $\Gamma_p = 2$–5 pmol cm^{-2} is 0.2–0.5% of the close-packed limit.

The following assumptions underlie this section:

1. The probe-thiol adsorbs to gold through the thiol–gold bond; MCH backfilling fills the remaining gold surface.
2. $\Gamma_{\max} = 930$ pmol cm^{-2} is an upper bound on MCH packing; the actual MCH density is lower because probe-thiols occupy a fraction of the surface.

C6FC reference reporter and ratiometric signal

C6FC (Fc–$(CH_2)_6$–SH) is co-immobilised with the probe-thiol at a molar ratio that is to be optimised on the fabricated electrode. Its function is threefold.

1. **Internal area reference.** The Fc signal I_{Fc} is proportional to the number of C6FC molecules on the electrode and is insensitive to MMP-9 activity, because C6FC contains no peptide substrate. I_{Fc} therefore tracks electrode area and surface loading, providing a normalisation factor that cancels run-to-run variation in electrode geometry or deposition yield.

2. **Fouling correction.** Non-specific protein adsorption attenuates both the MB and Fc signals by approximately the same factor (fouling layer acts as a diffusion barrier). Taking the ratio $I_{\mathrm{MB}}/I_{\mathrm{Fc}}$ suppresses this common-mode attenuation.
3. **Ratiometric readout.** The device signal is the ratio:

$$R(t) = \frac{I_{\mathrm{MB}}(t)}{I_{\mathrm{Fc}}} \tag{I.2}$$

where I_{Fc} is measured simultaneously with $I_{\mathrm{MB}}(t)$ in the same SWV scan. The normalised fractional cleavage is:

$$\frac{\Delta R}{R_0} = \frac{R_0 - R(t)}{R_0} = f_{\mathrm{cleaved}}(t) \tag{I.3}$$

provided I_{Fc} is stable over the assay window (see assumption below).

The electron-transfer kinetics of Fc-terminated alkanethiol chains on gold are characterised in Smalley et al. (2003) [8]: the tunnelling decay constant through the alkyl bridge is $\beta = 0.94 \pm 0.06$ Å^{-1}, confirming that ET from Fc to gold through the C_6 chain is fast relative to the SWV timescale and produces a sharp, well-defined peak. The same β value applies to the MB probe-thiol chain (Section IV).

The following assumption underlies this subsection. MMP-9 cleaves SGKIPRTLTA but has no known activity against the Fc–$(CH_2)_6$–SH molecule, which carries no peptide substrate; I_{Fc} is therefore expected to be stable over the 60-minute assay window. This has not been confirmed by a control experiment on the fabricated electrode and should be verified before ratiometric calibration is finalised. The target C6FC surface density relative to probe-thiol density is not confirmed from any paper in `papers/layer_07/` and must be optimised empirically.

II. Deposition Protocol and Target Surface Density

Confirmed deposition protocol

Wcisło et al. (2021) [11] describe immobilisation of a cysteine-terminated peptide substrate (SH–$(CH_2)_{10}$–CO–Arg-Val-Arg-Arg) on gold by the two-step procedure applicable to this device:

1. Immerse polished gold electrode in 5 mM degassed ethanolic solution of the peptide-thiol for 20 h at room temperature.
2. Rinse with ethanol and deionised water; dry under N_2.
3. Backfill with 2 mM ethanolic 1-octanethiol (analogous to MCH) for 60 min, to eliminate pinholes and block non-specific adsorption.
4. Optional protein block (BSA, 1% in PBS, 60 min at 4°C) if wound-matrix fouling is anticipated.

For this device, 1-octanethiol is replaced by MCH (6-mercapto-1-hexanol, HS–$(CH_2)_6$–OH) to match the C_6 alkyl chain length of the probe-thiol spacer. The target deposition concentration (5 mM) and time (20 h) are adopted from Wcisło 2021 [11]; exact optimisation for the specific SGKIPRTLTA–MB probe must be confirmed by measuring Γ_p on the fabricated electrode.

Langmuir adsorption framework

The equilibrium coverage of a thiol on gold from a solution of molar concentration c is described by the Langmuir isotherm:

$$\Gamma_p = \Gamma_{\max} \cdot \frac{K_L\, c}{1 + K_L\, c} \tag{II.1}$$

where $K_L = k_{\mathrm{ads}}/k_{\mathrm{des}}$ (M^{-1}) is the Langmuir adsorption constant and $\Gamma_{\max}$ is the monolayer saturation coverage.

Saturation coverage. For the directly analogous Fc-tetrapeptide-PEG_7 system on gold (PEG spacer, short peptide sequence, Fc label, MCH backfill), Adjémian 2010 [12] report saturation at $\Gamma_{\max} \approx 6 \times 10^{-11}$ mol cm^{-2} = 60 pmol cm^{-2} (Experimental Section, p. 10349 of [12]). This is substantially lower than the alkanethiol limit (930 pmol cm^{-2}, Eq. (I.1)) because the bulky PEG–peptide headgroup sterically limits packing. This value is used as an estimate of $\Gamma_{\max}$ for the SGKIPRTLTA–MB probe, pending direct coulometric measurement on the fabricated electrode.

Effectively irreversible adsorption. Thiol adsorption on gold is effectively irreversible under typical deposition conditions: the desorption rate constant k_{des} is negligible on the timescale of SAM preparation, giving $K_L \gg 1/c$ at any practical concentration. The Langmuir isotherm (Eq. (II.1)) therefore sits on its saturating plateau ($\Gamma_p \approx \Gamma_{\max}$) at both the micromolar concentrations used in Adjémian 2010 [12] and the 5 mM concentration adopted from Wcisło 2021 [11]. Coverage is consequently governed by *kinetic* Langmuir rather than equilibrium concentration:

$$\frac{d\Gamma_p}{dt} = k_{\text{ads}}\, c\, (\Gamma_{\max} - \Gamma_p) \tag{II.2}$$

The solution is $\Gamma_p(t) = \Gamma_{\max}(1 - e^{-k_{\text{ads}} c t})$: coverage increases toward $\Gamma_{\max}$ at a rate set by exposure time, not by the position of the equilibrium isotherm. This is precisely the regime exploited by Adjémian 2010, who achieve controlled sub-saturating coverages ($\Gamma_p \approx 0.4$–2×10^{-11} mol cm^{-2}) by limiting the immersion time to 30–120 s at 2–5 µM [12], rather than by adjusting concentration. The Wcisło 20 h protocol at 5 mM drives the system fully to $\Gamma_{\max}$; working coverage well below saturation must therefore be achieved by controlling immersion time, not by relying on equilibrium partitioning.

Working coverage selection. The device requires $\Gamma_p = 2$–5 pmol cm^{-2} (enzyme-access constraint, Section II), substantially below $\Gamma_{\max} = 60$ pmol cm^{-2}. Equation (II.2) implies this is reached at short exposure times (or very dilute probe concentration), consistent with the Adjémian 2010 approach. The exact immersion time that yields $\Gamma_p = 2$–5 pmol cm^{-2} for the SGKIPRTLTA–MB probe must be calibrated on the fabricated electrode by coulometric integration of the reductive desorption charge [6].

Three-component competitive Langmuir. With probe-thiol, C6FC, and MCH all present in the deposition solution, the three species compete for gold sites. The competitive Langmuir isotherm for component i is:

$$\theta_i = \frac{K_{L,i}\, c_i}{1 + K_{L,\text{probe}}\, c_{\text{probe}} + K_{L,\text{C6FC}}\, c_{\text{C6FC}} + K_{L,\text{MCH}}\, c_{\text{MCH}}} \tag{II.3}$$

The coverage ratio $\Gamma_{\text{probe}}/\Gamma_{\text{C6FC}}$ therefore depends on $K_{L,\text{probe}}\, c_{\text{probe}} \,/\, K_{L,\text{C6FC}}\, c_{\text{C6FC}}$. The individual Langmuir constants $K_{L,\text{probe}}$, $K_{L,\text{C6FC}}$, and $K_{L,\text{MCH}}$ for the specific thiols used here are not confirmed from any paper in `papers/layer_07/`; the deposition solution molar ratios must therefore be optimised empirically. Once deposited, the individual surface densities Γ_{probe} and Γ_{C6FC} should be verified by reductive desorption coulometry (Widrig [6]) or by integrating the separate CV peaks of MB and Fc in the same post-deposition scan.

Confirmed functional coverage limits

Adjémian et al. (2010) [12] characterised the relationship between surface coverage Γ_0 and cleavage efficiency for a ferrocene (Fc)-labelled tetrapeptide end-grafted on gold via a PEG_7 linker, using MCH backfill (architecture directly analogous to the MB-SGKIPRTLTA–PEG_6 probe). Their key finding (Fig. 2 and text):

- At saturated undiluted layers ($\Gamma_0 \approx 6 \times 10^{-11}$ mol cm^{-2} = 60 pmol cm^{-2}), cleavage yield drops to ∼50% because chains form compact domains inaccessible to the enzyme.
- For $\Gamma_0 \leq 10^{-11}$ mol cm^{-2} = 10 pmol cm^{-2} with MCH backfill, cleavage yield exceeds 80%.
- The working coverage used throughout Adjémian 2010 is $(0.5$–$1) \times 10^{-11}$ mol cm^{-2} = 5–10 pmol cm^{-2}.

The device target $\Gamma_p = 2$–5 pmol cm^{-2} therefore sits within the confirmed functional regime. Adjémian 2010 also confirms that surface kinetics follow the Michaelis–Menten two-step mechanism (Layer 9), with enzymatic constants measurable directly from the electrode response.

Parameter	**Status**	**Source**
Deposition protocol	Confirmed (5 mM EtOH, 20 h, MCH backfill)	Wcisło 2021 [11]
Functional Γ_p limit	Confirmed (≤ 10 pmol cm^{-2})	Adjémian 2010 [12]
k_{cat}, K_m for SGKIPRTLTA	Confirmed (86.4 s^{-1}, 1.3 mM)	Kridel 2001 [10]
Absolute Γ_p on fabricated electrode	**Unconfirmed**: must be measured	Coulometric integration
Surface $k_{\text{eff,surf}}$	**Unconfirmed**: must be measured on device	Layer 9

Table 1: Status of probe-loading parameters. The two unconfirmed items require measurements on the fabricated electrode; they cannot be borrowed from solution or analogous probes.

Probe spacing constraint from enzyme size

Once Γ_p is established from the literature, the average probe spacing is:

$$\ell = (\Gamma_p \cdot N_A)^{-1/2} \tag{II.4}$$

For the sensor to function, MMP-9 (molecular weight 92 kDa, hydrodynamic radius ≈ 3–4 nm) must approach and bind each surface-immobilised peptide. This imposes the requirement $\ell \gtrsim 2 \times r_{\text{MMP-9}} \approx 6$–$8$ nm, corresponding to $\Gamma_p \lesssim 3$ pmol cm^{-2}. At higher probe densities, neighbouring probe molecules sterically block enzyme access, reducing the effective cleavage rate and underestimating free MMP-9 activity. This constraint must be confirmed with a cleavage-efficiency vs. coverage measurement on the fabricated electrode.

The following assumptions underlie this section:

1. Probe-thiol adsorbs to gold via the thiol–gold bond; MCH backfilling fills remaining surface sites.
2. A target of $\Gamma_p \lesssim 3$ pmol cm^{-2} is adopted on steric grounds (enzyme-access requirement); this value is not confirmed from any paper in `papers/layer_07/` and is flagged as Limitation 1.

III. SAM Stability over the Assay Window

Three mechanisms can degrade the SAM during the 60-minute assay: oxidative desorption by H_2O_2, thiol exchange with free thiols in the wound matrix, and electrochemical desorption by the SWV scan potential.

Applicability of alkanethiol literature. All three subsections draw on papers that measured simple n-alkanethiol SAMs: hydrophobic chains with no polar headgroups, minimal solvent exposure, and compact crystalline packing on gold. The probe-thiol in this device is not a simple alkanethiol: it carries a PEG_6 polyether linker and a 10-residue charged peptide headgroup, both hydrophilic and solvent-swollen at pH 7.4. The MCH diluent (HS–$(CH_2)_6$–OH, $n = 6$) is structurally closer to the literature systems but is interdigitated with the bulky probe-thiol at 2–5 pmol cm^{-2}, which disrupts the close-packed alkanethiol geometry on which published rate constants and desorption potentials were measured. The alkanethiol data below provide order-of-magnitude bounds only and are not quantitatively applicable to this mixed peptide-thiol/MCH monolayer without experimental confirmation on the fabricated electrode.

Oxidative desorption

Schoenfisch and Pemberton characterised alkanethiol SAM oxidation under ambient conditions [5]. Their central finding is that ozone (O_3), not O_2 or UV, is the primary oxidant in air, converting Au–SR to Au^0 + RSO_xH. Under ambient O_3 at 28–60 ppb, a C_{12} SAM on gold lost 36.9% of its total sulphur in 1 h (Table 5 of [5]).

In the present device, the oxidant is H_2O_2 at 10–100 μM in aqueous wound fluid; ozone is absent from the aqueous phase. The relative oxidising power of the two species is reflected in their standard reduction potentials:

$$E^\circ(O_3/O_2) = +2.07 \text{ V}, \qquad E^\circ(H_2O_2/H_2O) = +1.76 \text{ V} \tag{III.1}$$

The 310 mV thermodynamic advantage of O_3 corresponds to a rate advantage of several orders of magnitude for thiolate oxidation. The Schoenfisch data therefore provide a conservative upper bound: even under the harshest oxidative conditions tested (ambient O_3 in air), SAM loss is approximately 37% in 1 h for a C_{12} chain. Aqueous H_2O_2 at wound-fluid concentrations is a substantially weaker oxidant; oxidative SAM loss over the 60-minute assay is not expected to approach this bound.

No quantitative rate constant k_{ox} for H_2O_2 oxidation of thiol SAMs in solution is available in `papers/layer_07/`. The argument above is comparative only; see Limitation 2.

Thiol exchange

Schlenoff et al. investigated the exchange of adsorbed alkanethiols on gold by free thiol in solution [7]. The key finding is that the exchange rate shows "very little dependence of rate on thiol concentration," meaning the process is not bimolecular but involves a rate-limiting surface step. In hexane, the pseudo-first-order exchange rate constant is [7]:

$$k_1 = 1.7 \times 10^{-5} \text{ s}^{-1} \quad \text{(hexane, 22°C)} \tag{III.2}$$

The fraction exchanged at time t is $f = 1 - \exp(-k_1 t)$. At $t = 3600$ s (60 min):

$$f = 1 - \exp(-1.7 \times 10^{-5} \times 3600) = 1 - e^{-0.061} \approx 5.9\% \tag{III.3}$$

Equation (III.3) is a strict upper bound: Schlenoff et al. further show that exchange in aqueous conditions is negligible, with essentially no exchange detected even after 125 h in water (Figures 3–6 of [7]). They conclude that desorption is "least pronounced in aqueous systems." Wound fluid is an aqueous matrix; thiol exchange does not contribute meaningfully to SAM loss over the assay window.

Electrochemical desorption

Reductive desorption of alkanethiol SAMs on gold occurs via:

$$\text{Au–SR} + e^- \longrightarrow \text{Au}^0 + \text{RS}^- \tag{III.4}$$

and is only possible when the electrode potential reaches or exceeds the desorption potential E_{des}. Widrig et al. measured E_{des} for a series of ω-alkanethiol SAMs on gold by cyclic voltammetry in 0.5 M KOH, finding a linear dependence on carbon chain length n [6]:

$$E_{\text{des}} = (-0.83 - 0.020\,n)\ \text{V vs Ag/AgCl} \quad (0.5\ \text{M KOH}) \tag{III.5}$$

For MCH ($n = 6$):

$$E_{\text{des}} = -0.83 - 0.020 \times 6 = -0.83 - 0.12 = -0.95\ \text{V} \tag{III.6}$$

Love et al. report that desorption of alkanethiol SAMs from gold typically occurs in the range -1.0 ± 0.25 V vs Ag/AgCl [1], giving a conservative lower bound of -0.75 V. The SWV scan window runs from -0.65 V to $+0.1$ V (Layer 6). The margin between the scan minimum and the conservative lower bound is:

$$\Delta E = E_{\text{des,min}} - E_{\text{scan,min}} = -0.75\ \text{V} - (-0.65\ \text{V}) = -0.10\ \text{V} \tag{III.7}$$

The scan potential is 100 mV more positive than the most conservative desorption estimate; the Widrig value for MCH is 300 mV more negative than the scan limit. Electrochemical desorption does not occur during the assay.

The foregoing analysis uses Widrig's formula and Love's range, both derived for simple n-alkanethiol SAMs whose hydrophobic chains have minimal solvent penetration and desorb as a sharp stripping peak from a well-ordered layer. The probe-thiol (HS–$(CH_2)_6$–PEG_6–SGKIPRTLTA–MB) and MCH have significant hydrophilic character: the PEG_6 block is a polyether that binds water, and the peptide headgroup bears partial charges at pH 7.4. A solvated, structurally disordered layer has different desorption thermodynamics: the departing thiolate is stabilised by water, lowering the thermodynamic barrier relative to a hydrophobic chain. The actual E_{des} for the probe-thiol/MCH monolayer may therefore be less negative than the Widrig and Love values indicate. This is an unconfirmed transferability issue; see Limitation 4.

The following assumptions underlie this section:

1. The oxidative stability argument uses the Schoenfisch O_3/air benchmark as an upper bound; no quantitative H_2O_2 rate constant is claimed (Limitation 2).
2. Equation (III.2) and the 5.9% bound apply in organic solvent; aqueous exchange is negligible per Schlenoff [7] (Limitation 3).
3. The Love range [1] is used as the primary reference for device-condition E_{des}; the Widrig formula (equation III.5) is supporting data measured specifically in alkaline electrolyte (Limitation 4).

IV. Ca^{2+} Artefact and Ratiometric Cancellation

Mechanism

Wound fluid contains ionised Ca^{2+} at a concentration comparable to serum (∼1–2 mM; no paper in `papers/layer_07/` reports a quantitative wound-fluid Ca^{2+} value: this is an unconfirmed estimate used for the sensitivity calculation below). If the probe-thiol architecture includes a PEG spacer between the alkyl chain and the peptide, Ca^{2+} can coordinate the polyether oxygen atoms in a crown-ether-like interaction, contracting the spacer and decreasing the probe-to-gold bridge length by Δd. Electron transfer (ET) from MB to gold through the bridge follows a tunnelling mechanism, so a decrease in bridge length raises the apparent ET rate constant k_{ET} and increases the SWV peak current of *uncleaved* probes.

For the peptide cleavage sensor (signal-off: MB released on cleavage), this Ca^{2+}-induced increase in k_{ET} acts on the uncleaved probe population equally before (I_0) and after ($I(t)$) the assay. The measured quantity is:

$$\frac{\Delta I}{I_0} = \frac{I_0 - I(t)}{I_0} \tag{IV.1}$$

If Ca^{2+} multiplies k_{ET} by a factor R for all uncleaved probes identically, then $I_0 \to R\,I_0$ and $I(t) \to R\,(1-f)\,I_0$ where f is the fraction cleaved. Substituting:

$$\frac{\Delta I}{I_0} = \frac{R\,I_0 - R(1-f)\,I_0}{R\,I_0} = f \tag{IV.2}$$

The Ca^{2+} factor R cancels exactly in the signal-off ratio, provided both I_0 and $I(t)$ are measured in the same buffer (same Ca^{2+} concentration). The signal is inherently self-normalising against Ca^{2+}-induced changes in electron transfer rate, as long as the measurement protocol does not change the buffer between baseline and post-incubation reads.

Tunnelling decay constant

The ET rate constant through a molecular bridge of length d is:

$$k_{ET} = k_{ET}^0 \exp(-\beta\, d) \tag{IV.3}$$

where β is the tunnelling decay constant. Smalley et al. measured β for ferrocene-terminated (Fc) alkanethiol chains tethered to gold in 1.0 M $HClO_4$, by fitting the chain-length dependence of the standard rate constant k° [8]:

$$\beta = 0.94 \pm 0.06 \text{ Å}^{-1} = (9.4 \pm 0.6) \times 10^9 \text{ m}^{-1} \tag{IV.4}$$

This measurement used ferrocene (Fc) as the redox probe, while the present device uses methylene blue (MB). The value of β is adopted here because β is a property of the alkyl bridge that electrons tunnel through, not of the terminal redox group. Smalley et al. note that $\beta \approx 1.0$ Å^{-1} is consistent across multiple redox probes studied through the same alkyl bridge [8]. The Fc-derived value therefore transfers directly to the MB system.

Sensitivity of k_{ET} to bridge contraction

If Ca^{2+} coordination contracts the PEG spacer by Δd, the fold increase in k_{ET} follows from equation (IV.3):

$$\frac{k_{ET}^{Ca}}{k_{ET}^0} = \exp(\beta\,\Delta d) \tag{IV.5}$$

For a representative contraction of $\Delta d = 0.2$ nm (one carbon–carbon bond displacement, used as a sensitivity bound since Δd is not measured for this architecture; see Limitation 5):

$$\begin{aligned} \frac{k_{ET}^{Ca}}{k_{ET}^0} &= \exp(9.4 \times 10^9 \text{ m}^{-1} \times 0.2 \times 10^{-9} \text{ m}) \\ &= \exp(1.88) \approx 6.5 \end{aligned} \tag{IV.6}$$

Both the active and control electrodes carry the same probe-thiol/MCH SAM and are exposed simultaneously to the same wound fluid, so this fold increase applies identically to both.

Residual error from probe density change after cleavage

Equation (IV.2) assumes that Ca^{2+} affects uncleaved probes with the same k_{ET} before and after the assay. This holds exactly if the cleaved probe stub (the fragment remaining on the electrode after MB release) has negligible electron transfer compared with the intact probe. If the stub retains some residual MB (partial cleavage) or if cleavage changes the spacer geometry in a way that modifies the Ca^{2+} response of neighbouring probes, a second-order residual arises. The magnitude of this residual depends on the specific peptide architecture and is not quantifiable without a calibration measurement. It is expected to be small (second-order in f and in $\delta k_{ET,stub}$), but experimental confirmation that $\Delta I/I_0$ is Ca^{2+}-independent in the assay buffer is recommended.

Limitations of this model

1. **The deposition protocol and functional coverage limit are now confirmed; the absolute Γ_p on the fabricated electrode must still be measured.** Wcisło 2021 [11] provides a confirmed peptide-thiol deposition protocol (5 mM ethanolic solution, 20 h, MCH backfill). Adjémian 2010 [12] confirms the functional coverage regime ($\Gamma_0 \leq 10$ pmol cm^{-2} for >80% cleavage yield with MCH backfill). The device target of 2–5 pmol cm^{-2} is within this regime. What remains unconfirmed is the *actual* Γ_p achieved on the fabricated SGKIPRTLTA–MB electrode under the adopted deposition conditions. This must be measured directly by reductive desorption coulometry (method: Widrig [6]) or equivalent before Layer 8 and Layer 9 outputs can be quantified.
2. **No quantitative k_{ox} for H_2O_2 oxidation of thiol SAMs.** Schoenfisch and Pemberton [5] measure SAM oxidation by O_3 in air, not by H_2O_2 in solution. No paper in `papers/layer_07/` provides k_{ox} for aqueous H_2O_2. The oxidative stability argument is comparative only; experimental confirmation of SAM integrity after 60 min in wound-fluid H_2O_2 is required before the assay window can be certified.
3. **Thiol exchange rate confirmed in hexane only.** Schlenoff et al. [7] measure $k_1 = 1.7 \times 10^{-5}$ s^{-1} in hexane and demonstrate negligible aqueous exchange qualitatively; a quantitative rate constant for exchange by Cys or GSH in phosphate-buffered wound fluid at pH 7.4 is not available.
4. **E_{des} measured in 0.5 M KOH on hydrophobic alkanethiols; both conditions differ from this device.** The Widrig formula (equation III.5) was determined at pH ≈ 13.7 [6]. At wound-fluid pH 7.4 the Nernst equation predicts that E_{des} shifts approximately $+(13.7 - 7.4) \times 0.059 \approx +0.37$ V toward positive values (because RS^- is protonated at pH 7.4), placing E_{des} near $-0.95 + 0.37 = -0.58$ V. This is only 70 mV positive of the scan minimum of −0.65 V, compared with the 300 mV margin in alkaline electrolyte. The Love range [1] (-1.0 ± 0.25 V, conditions unspecified) is used as the primary working reference. A second applicability concern arises from molecular structure: Widrig's formula was measured on hydrophobic *n*-alkanethiols with no polar headgroups and no solvent penetration into the SAM interior. The probe-thiol and MCH in this device have substantial hydrophilic character (PEG_6 polyether, peptide headgroup, hydroxyl terminus of MCH). Greater water exposure and structural disorder lower the thermodynamic barrier to desorption and may shift E_{des} toward less negative values relative to the Widrig formula. The formula is not transferable to this mixed peptide-thiol/MCH monolayer; experimental determination of E_{des} in the assay buffer on the fabricated electrode is required before the scan window is certified safe.
5. **Ca^{2+} self-cancellation in $\Delta I/I_0$ is conditional on the measurement protocol.** Equation (IV.2) shows that the Ca^{2+} artefact cancels in $\Delta I/I_0$ only if I_0 and $I(t)$ are measured in the same buffer (same Ca^{2+} concentration). If the baseline I_0 is measured before adding wound fluid (in calibration buffer without Ca^{2+}) and $I(t)$ is measured in wound fluid (with Ca^{2+}), the cancellation fails and a spurious signal arises. Protocol design must ensure that both measurements are in the same ionic environment. Whether a PEG spacer is present and its Ca^{2+} affinity are also unconfirmed for this probe architecture.

References

All papers in `electrochemical_device/papers/layer_07/`.

[1] Love JC, Estroff LA, Kriebel JK, Nuzzo RG, Whitesides GM (2005). Self-assembled monolayers of thiolates on metals as a form of nanotechnology. *Chem Rev* 105(4):1103–1169. **love2005.pdf** – Γ_{max} cross-check (≈ 750 pmol cm^{-2}, p. 1108), $E_{des} = -1.0 \pm 0.25$ V (p. 1122) confirmed from text.

[2] Herne TM, Tarlov MJ (1997). Characterization of DNA probes immobilized on gold surfaces. *J Am Chem Soc* 119(38):8916–8920. **herne1997.pdf** – *DNA-thiol specific. KH_2PO_4 buffer requirement, log-time deposition kinetics, and MCH displacement of non-thiolated DNA all pertain to HS-ssDNA with 25 charged phosphate groups per strand. These results do not transfer to the peptide-thiol probe and are not used for deposition protocol in this layer.*

[3] Steel AB, Herne TM, Tarlov MJ (1998). Electrochemical quantitation of DNA immobilized on gold. *Anal Chem* 70(22):4670–4677. **steel1998.pdf** – *DNA-thiol specific. The functional coverage limit (4×10^{12} molecules cm^{-2}) was determined by hybridisation steric hindrance. Enzyme-cleavage steric limits for MMP-9 are different and not covered by this paper.*

[4] Petrovykh DY, Kimura-Suda H, Whitman LJ, Tarlov MJ (2003). Quantitative analysis and characterization of DNA immobilized on gold. *J Am Chem Soc* 125(17):5219–5226. **petrovykh2003.pdf** – *DNA-thiol specific. Log-linear adsorption kinetics and the non-Langmuir conclusion apply to HS-ssDNA. Peptide-thiol adsorption kinetics are not covered by this paper.*

[5] Schoenfisch MH, Pemberton JE (1998). Air stability of alkanethiol self-assembled monolayers on silver and gold surfaces. *J Am Chem Soc* 120(18):4502–4513. **schoenfisch1998.pdf** – O_3 as primary oxidant confirmed from text; C_{12} SAM 36.9% sulphur loss in 1 h confirmed from Table 5. Paper does not give k_{ox} for aqueous H_2O_2; value attributed to this paper in `sam_stability.py` is not confirmed and has been removed.

[6] Widrig CA, Chung C, Porter MD (1991). The electrochemical desorption of n-alkanethiol monolayers from polycrystalline Au and Ag electrodes. *J Electroanal Chem* 310(1–2):335–359. **widrig1991.pdf** – $\Gamma_{\max} = 9.3 \times 10^{-10}$ mol cm^{-2} (p. 351) confirmed; E_{des} formula (equation III.5) confirmed from text. Measured in 0.5 M KOH; pH correction noted in Limitation 4.

[7] Schlenoff JB, Li M, Ly H (1995). Stability and self-exchange in alkanethiol monolayers. *J Am Chem Soc* 117(50):12528–12536. **schlenoff1995.pdf** – $k_1 = 1.7 \times 10^{-5}$ s^{-1} in hexane confirmed from Table 3; concentration-independence of exchange rate confirmed from text; negligible aqueous exchange confirmed from Figures 3–6. Bimolecular rate model in `sam_stability.py` is not supported by this paper and has been removed.

[8] Smalley JF, Feldberg SW, Chidsey CED, Linford MR, Newton MD, Liu Y-P (2003). The kinetics of electron transfer through ferrocene-terminated alkanethiol monolayers on gold. *J Am Chem Soc* 125(7):2004–2013. **smalley2003.pdf** – $\beta = 0.94 \pm 0.06$ Å^{-1} confirmed from text (p. 2009). Measured using ferrocene (Fc) redox probe; value transfers to methylene-blue system because β is a property of the alkyl bridge, confirmed from discussion in [8].

[9] Becka AM, Miller CJ (1993). Electrochemistry at ω-hydroxy thiol coated electrodes. 3. Voltage independence of the electron tunneling barrier and measurements of redox kinetics at large overpotentials. *J Phys Chem* 97(23):6233–6239. **becka1993.pdf** – provides context for tunnelling through alkanethiol SAMs; the $\beta = 0.94$ Å^{-1} value used in this layer is from Smalley 2003 [8], not Becka 1993.

[10] Kridel SJ, Chen E, Kotra LP, Howard EW, Mobashery S, Smith JW (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J Biol Chem* 276(23):20572–20578. DOI: 10.1074/jbc.M100900200. **PIIS0021925819405115.pdf** – $k_{\text{cat}} = 86.4$ s^{-1}, $K_m = 1.3$ mM, $k_{\text{cat}}/K_m = 67{,}500$ M^{-1}s^{-1} for peptide A11 (SGKIPRTLTA) confirmed from Table II. These solution-phase values are an upper bound for the surface $k_{\text{eff,surf}}$.

[11] Wcisło A, Małuch I, Niedziałkowski P, Ossowski T, Prahl A (2021). Label-free electrochemical test of protease interaction with a peptide substrate modified gold electrode. *Chemosensors* 9(8):199. DOI: 10.3390/chemosensors9080199. **10.3390@chemosensors9080199.pdf** – deposition protocol for peptide-thiol on gold (5 mM EtOH, 20 h, backfill with alkanethiol, BSA block) confirmed from Section 2.3.

[12] Adjémian J, Anne A, Cauet G, Demaille C (2010). Cleavage-sensing redox peptide monolayers for the rapid measurement of the proteolytic activity of trypsin and α-thrombin enzymes. *Langmuir* 26(12):10347–10356. DOI: 10.1021/la100397g. **10.1021@la100397g.pdf** – cleavage yield >80% for $\Gamma_0 \leq 10$ pmol cm^{-2} with MCH backfill confirmed from Fig. 2 and text (p. 10351); working coverage (0.5–1) $\times 10^{-11}$ mol cm^{-2} confirmed. Architecture (Fc-peptide-PEG tethered on Au, MCH backfill) is directly analogous to this device.

[13] Sfragano PS, Moro G, Polo F, Palchetti I (2021). The role of peptides in the design of electrochemical biosensors for clinical diagnostics. *Biosensors* 11(8):246. DOI: 10.3390/bios11080246. **sfragano2021.pdf** – review of peptide-based electrochemical biosensors; confirms PEG linker strategies, cysteine-gold anchoring, and protease cleavage signal-off mechanisms consistent with this device architecture.